

$$\begin{array}{c}
\mathcal{K}_{0^+}^{\#1} \quad \mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{0^+}^{\#1} \dagger \quad \begin{array}{|c|c|} \hline -\frac{\Upsilon_1 k^2}{2} & 0 \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#1} \dagger^{\alpha\beta\chi} \quad \begin{array}{|c|c|} \hline 0 & 3k^2 \binom{(4)}{K_{15}} + 3 \binom{(2)}{K_1} \\ \hline \end{array} \quad \mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta} \quad \mathcal{K}_{1^+}^{\#1}{}_{\alpha} \quad \mathcal{K}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta} \quad \begin{array}{|c|c|c|c|} \hline \frac{1}{2} (\Upsilon_2 k^2 + 4 \binom{(2)}{K_1}) & \frac{\Upsilon_3 k^2 + 4 \binom{(2)}{K_1}}{2 \sqrt{2}} & 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta} \quad \begin{array}{|c|c|c|c|} \hline \frac{\Upsilon_3 k^2 + 4 \binom{(2)}{K_1}}{2 \sqrt{2}} & \frac{1}{2} (\Upsilon_4 k^2 + 2 \binom{(2)}{K_1}) & 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha} \quad \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha} \quad \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \quad \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta} \quad \begin{array}{|c|c|} \hline -\frac{\Upsilon_5 k^2}{2} & 0 \\ \hline \end{array} \\
\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi} \quad \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}$$

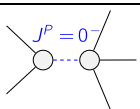
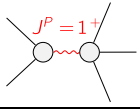
$$\begin{array}{c}
\mathcal{J}_{0^+}^{\#1} \quad \mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{0^+}^{\#1} \dagger \quad \begin{array}{|c|c|} \hline \frac{1}{\text{Det}(0^+)} & 0 \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#1} \dagger^{\alpha\beta\chi} \quad \begin{array}{|c|c|} \hline 0 & \frac{1}{3(k^2 \binom{(4)}{K_{15}} + \binom{(2)}{K_1})} \\ \hline \end{array} \quad \mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta} \quad \mathcal{J}_{1^+}^{\#1}{}_{\alpha} \quad \mathcal{J}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta} \quad \begin{array}{|c|c|c|c|} \hline \frac{\Upsilon_4 k^2 + 2 \binom{(2)}{K_1}}{2 \text{Det}(1^+)} & \frac{-\Upsilon_3 k^2 - 4 \binom{(2)}{K_1}}{2 \sqrt{2} \text{Det}(1^+)} & 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta} \quad \begin{array}{|c|c|c|c|} \hline \frac{-\Upsilon_3 k^2 - 4 \binom{(2)}{K_1}}{2 \sqrt{2} \text{Det}(1^+)} & \frac{\Upsilon_2 k^2 + 4 \binom{(2)}{K_1}}{2 \text{Det}(1^+)} & 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha} \quad \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha} \quad \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \quad \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta} \quad \begin{array}{|c|c|} \hline \frac{1}{\text{Det}(2^+)} & 0 \\ \hline \end{array} \\
\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta\chi} \quad \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}$$

Abbreviations used in matrices

$$\begin{aligned}
\Upsilon_1 &= 3 \binom{(4)}{K_1} + \binom{(4)}{K_3} + \binom{(4)}{K_6} \& \& \Upsilon_2 = 4 \binom{(4)}{K_{15}} - \binom{(4)}{K_3} + \binom{(4)}{K_6} \& \& \Upsilon_3 = 4 \binom{(4)}{K_{15}} + \binom{(4)}{K_4} \& \& \\
\Upsilon_4 &= 2 \binom{(4)}{K_{15}} + \binom{(4)}{K_3} + \binom{(4)}{K_4} \& \& \Upsilon_5 = \binom{(4)}{K_3} + \binom{(4)}{K_6} \& \& \text{Det}(0^+) = -\frac{1}{2} k^2 (3 \binom{(4)}{K_1} + \binom{(4)}{K_3} + \binom{(4)}{K_6}) \& \& \\
\text{Det}(1^+) &= \frac{1}{8} (k^4 (-2 \binom{(4)}{K_3}^2 - 2 \binom{(4)}{K_3} \binom{(4)}{K_4} - \binom{(4)}{K_4}^2 + 2 (\binom{(4)}{K_3} + \binom{(4)}{K_4}) \binom{(4)}{K_6} + 4 \binom{(4)}{K_{15}} (\binom{(4)}{K_3} + \binom{(4)}{K_6})) + 4 k^2 (\binom{(4)}{K_3} + \binom{(4)}{K_6}) \binom{(2)}{K_1}) \& \& \\
\text{Det}(2^+) &= -\frac{1}{2} k^2 (\binom{(4)}{K_3} + \binom{(4)}{K_6})
\end{aligned}$$

Lagrangian

$$\begin{aligned}
& \binom{(2)}{K_1} \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} - 2 \binom{(2)}{K_1} \mathcal{K}^{\alpha\beta\chi} \mathcal{K}_{\beta\alpha\chi} + \binom{(4)}{K_1} \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta + \\
& \binom{(4)}{K_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\alpha\chi} + \binom{(4)}{K_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \binom{(4)}{K_6} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} + \frac{1}{2} \binom{(4)}{K_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \\
& \frac{1}{2} \binom{(4)}{K_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \binom{(4)}{K_{15}} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \binom{(4)}{K_{15}} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_1^{\#2\alpha} = 0$	3	$\partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi} = 0$	
$\mathcal{J}_1^{\#1\alpha} = 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta_{\beta}{}^\chi + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}_{\beta} = \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}_2^{\#1\alpha\beta\chi} = 0$	5	$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} +$ $4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}_{\delta}{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\alpha}_{\delta} =$ $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} +$ $2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta}_{\delta}{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\beta}_{\delta}$	
Total # constraint(s):	11		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$-\frac{\binom{(2)}{K_1}}{\binom{(4)}{K_{15}}}$	$-\frac{1}{3 \binom{(4)}{K_{15}}}$
	3	$\frac{4 (\binom{(4)}{K_3} + \binom{(4)}{K_6}) \binom{(2)}{K_1}}{2 \binom{(4)}{K_3}^2 + 2 \binom{(4)}{K_3} \binom{(4)}{K_4} + \binom{(4)}{K_4}^2 - 2 (\binom{(4)}{K_3} + \binom{(4)}{K_4}) \binom{(4)}{K_6} - 4 \binom{(4)}{K_{15}} (\binom{(4)}{K_3} + \binom{(4)}{K_6})}$	$-\frac{2 (\binom{(4)}{K_3}^2 + 8 \binom{(4)}{K_3} \binom{(4)}{K_4} + 3 \binom{(4)}{K_4}^2 - 4 \binom{(4)}{K_3} \binom{(4)}{K_6} - 4 \binom{(4)}{K_4} \binom{(4)}{K_6} + 2 \binom{(4)}{K_6}^2)}{(\binom{(4)}{K_3} + \binom{(4)}{K_6}) (2 \binom{(4)}{K_3}^2 + \binom{(4)}{K_4} (\binom{(4)}{K_4} - 2 \binom{(4)}{K_6}) + 2 \binom{(4)}{K_3} (\binom{(4)}{K_4} - \binom{(4)}{K_6}) - 4 \binom{(4)}{K_{15}} (\binom{(4)}{K_3} + \binom{(4)}{K_6}))}$
Resolved unitarity condition(s):		(Demonstrably impossible)	